**a** MERRA-2

**b** CTRL

**c** CTRL−MERRA-2

**d** $HI_{gbl}$

**e** $HI_{gbl}$−MERRA-2

**f** $HI_{mex}$

**g** $HI_{mex}$−MERRA-2

60°N
50°N
40°N
30°N
20°N

130°W 120°W 110°W 100°W 90°W

5500 5625 5750 5875 6000
$\Phi_{500}$ Climatology [m]

−100 −50 0 50 100
$\Phi_{500}$ Model Bias [m]